Genesis Forge - All Plans & Decisions

Handoff for another chat. Everything decided in the 2026-11-05/06 sessions, the approved build plan, and the concept images. **Nothing below is built yet** unless it says so.

1. About the mod

• Genesis Forge (GF): Zeven's custom NeoForge mod in CurseForge modpack instance 5423.
• Minecraft 1.21.1, NeoForge 21.1.262, Java 21, Mojang mappings, MixinExtras. Package **com.zeven.genesisforge**.
• Jar: **genesisforge-1.0.0.jar** (~26.8 MB), must stay under 30 MB. Path: C:\Users\USER\curseforge\minecraft\Instances\5423\mods\
• Main parts: Genesis Forge block + soul orbs, 4 spawner tiers, Soul Balls, Abyssal storage, Block Drill, Omni Nexus (9-tab station), Galaxium gear (tier above Netherite), 8 Galaxium mini-bosses, 13 apple trees, enchantment reworks.
• Enchant max levels (Omni Nexus): normal gear 10x vanilla max, Galaxium 50x.

Working rules (Zeven)

• Show concepts/previews and ask when numbers or requirements are unclear. No unreviewed design overhauls.
• Galaxium textures: no seams, tiling lines or mirrored faces. Designs: smooth, organic curves.
• Always say what was only compiled/tested vs. actually tested in the game client.
• Never replace the jar while Minecraft is running. Back up before installing.

2. Current status

• The mod's source code is lost (old /tmp/gf project gone). Plan: rebuild source from the jar (see section 9).
• Waiting for Zeven to upload **genesisforge-1.0.0.jar**.
• Build environment ready: a personal cloud environment 'supreme-couscous' allows the NeoForge + Mojang servers; an empty NeoForge 1.21.1 project (ModDevGradle 2.0.148) compiles in repo Zeven0X/supreme-couscous, folder genesisforge/.
• All decisions are also in that repo: docs/enchantment-rework-plan.md, docs/session-2026-11-05.md, docs/concepts/.

3. Bug found and in-game test results

BUG: Multishot stops at level 5 on the Galaxium Crossbow. Must follow the 50x rule = **50**. Other enchantments may have the same wrong cap: audit all 42 when the jar is available.

Test	Result
Giant 2x2 / mega 3x3 trees	Works, but bone meal amount must change (section 4)
Apples dropping under big trees (fruiting log)	Works
Spawner 24-spot line-of-sight	Works
Tier 3/4 spawner sneak on/off	Works
Dispensers firing Galaxium mini-boss eggs	NOT tested yet
Piercing on normal bows	Works
Omni Nexus brewing + 256-block beacon	Works

4. Tree bone meal change (decided)

Giant 2x2 = exactly 8x, mega 3x3 = exactly 24x the average bone meal ONE sapling of that type needs. Fixed count, not random. One sapling average = round(2 x slowness / 0.45).

Tree	1 sapling	Giant 2x2	Mega 3x3
Apple	9	72	216
Iron	13	104	312
Golden / Emerald / Diamond	22	176	528
Netherite	44	352	1,056
Galaxium	111	888	2,664
Enchanted trees	from own slowness	8x	24x

5. Gear tiers and level limit (decided)

• Normal gear 10x vanilla max, Galaxium 50x, plus two FUTURE gear sets (not created yet) at **75x** and **100x**.
• Minecraft 1.21.1 only stores enchant levels 1-255. **Decided: raise the limit above 255.** Risky: test in a world copy; check saving/reloading, /enchant, tooltips.
• Reworks are designed so their full value is reached on the 100x set.
• **Reach order:** Galaxium tools' +75% reach first, then the enchantment reach: 3 x 1.75 + 10 = 15.25 blocks.

6. Enchantment reworks (all decided, not built)

Max level columns: normal (10x) / Galaxium (50x) / 75x set / 100x set.

Enchantment	New effect	Normal	Galaxium	75x	100x
Smite	Every hit on ANY mob heals the attacker 0.005 HP/level. Keeps vanilla +2.5 dmg/level vs undead.	0.25 HP	1.25 HP	1.875 HP	2.5 HP
Knockback	+0.05 blocks reach/level (attack + blocks), held item only. Still pushes mobs. Adds after Galaxium +75%.	+1	+5	+7.5	+10
Silk Touch	+0.1 blocks reach/level (attack + blocks). Tools only, held item only, does NOT stack with Knockback.	+1	+5	+7.5	+10
Aqua Affinity	Helmet only. Fully underwater: 0.75%/level off the damage left AFTER all other reduction. Keeps faster underwater mining.	7.5%	37.5%	56.25%	75%
Impaling	+0.5 damage/level vs EVERY mob, thrown + melee. (Assumed: vanilla +2.5 vs aquatic kept.)	+25	+125	+187.5	+250
Punch	-0.4% projectile gravity/level, max 80%. Keeps knockback. Can also go on the Galaxium Crossbow (only).	8%	40%	60%	80%
Multishot	Vanilla +2 arrows/level, NO cap. From lvl 75: shoot 75+ degrees up -> arrow vanishes, ~1 s later one fast arrow falls on every target in 25 blocks (+1/level above 75). Full crossbow damage (Power, Flame, cosmic fire). Hits everyone except the shooter's tamed pets/companions.	21 arrows	101 arrows	rain 25 blocks	rain 50 blocks
Fortune	From lvl 200: auto-smelts all smeltable ores (copper, modded), Fortune multiplies, smelting XP. Ancient debris -> netherite scrap like diamond ore (~100 at lvl 200). Toggle in item settings; Silk Touch wins.	-	-	yes (225)	yes (300)
Luck of the Sea	From lvl 50 (Galaxium Fishing Rod): Nether lava fishing. Loot 10% each: blaze rod, magma cream, ghast tear, quartz, glowstone dust, gold nugget, netherite scrap, ancient debris; Netherite Apple sapling 2.5%, Enchanted Netherite Apple sapling 0.5%; 17% empty. Better odds at higher levels: redesign later.	-	yes	yes	yes
Fire Protection	Longer gap between ALL fire-related damage hits (fire, lava, Purple Cosmic Fire); hit size unchanged. Levels of 4 pieces add up, max at 1600.	1.3x	2.5x	3.25x	4x
Blast Protection	Extra explosion reduction, applied after all other reduction. 4 pieces add up, 90% at 1600.	9%	45%	67.5%	90%
Projectile Protection	Same as Blast, vs projectiles.	9%	45%	67.5%	90%
Feather Falling	+0.25 safe-fall blocks/level (boots). Adds to normal 3 + Galaxium Boots' 15 (max 118).	+10	+50	+75	+100

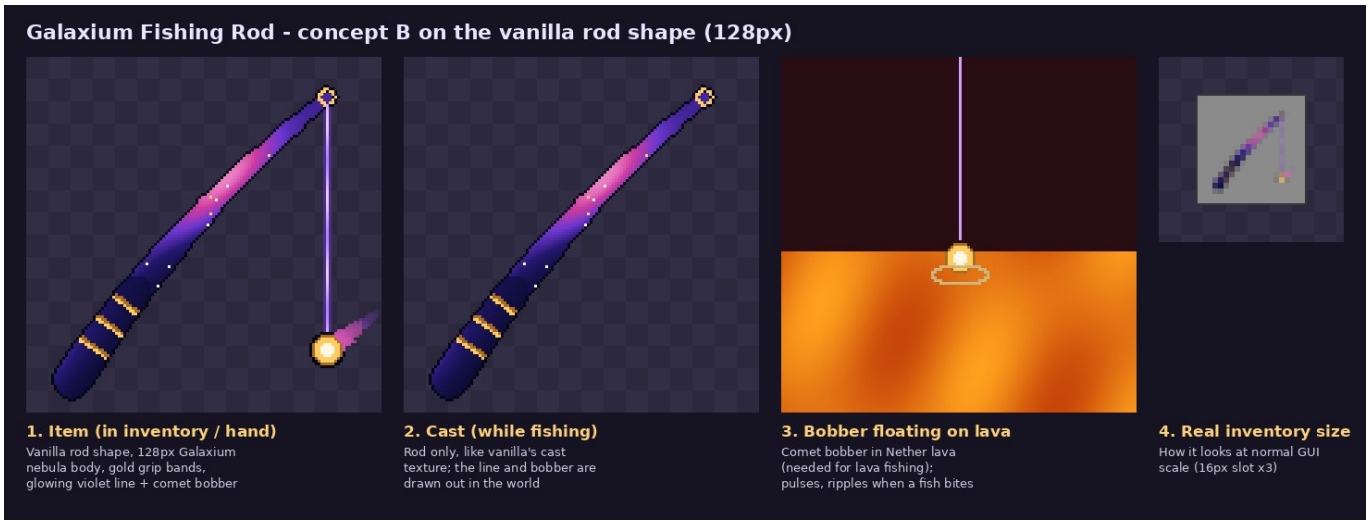
Protection columns are for a full armor set with the same level on every piece (one piece = a quarter). Fire/Blast/Projectile numbers assume linear scaling.

No rework / open

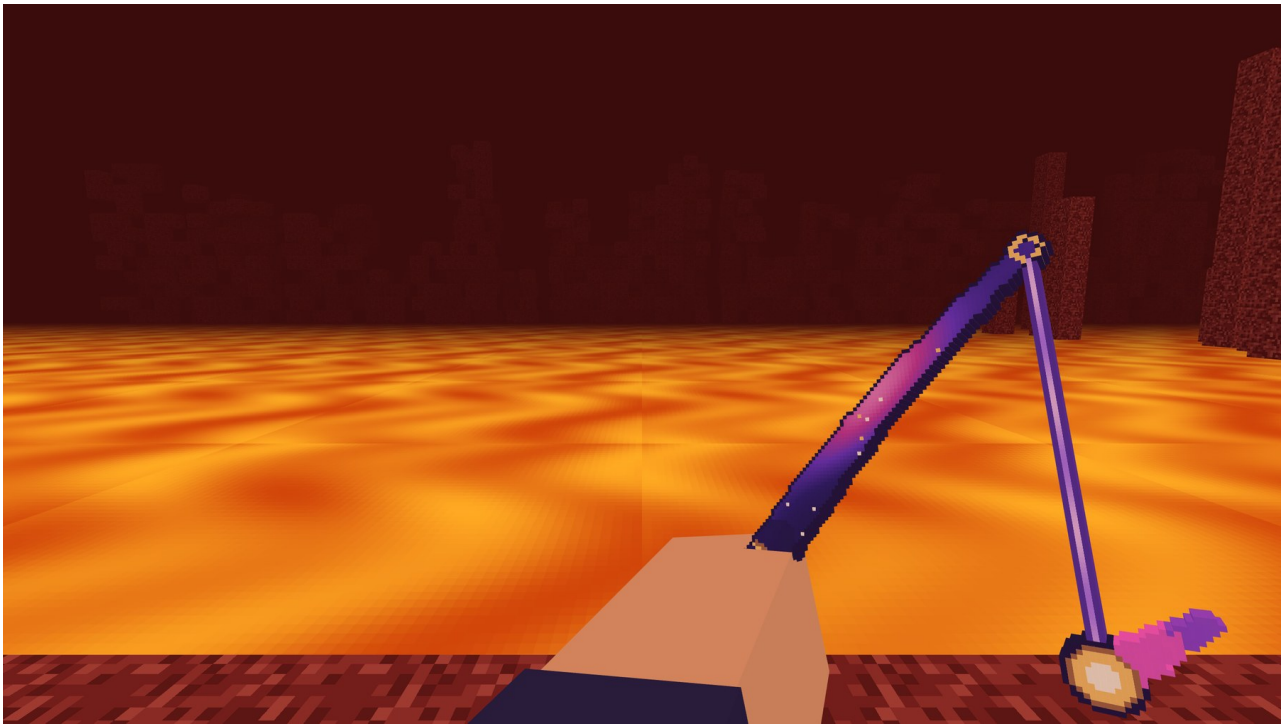
• No rework needed: Curse of Binding, Curse of Vanishing, Unbreaking, Density.
• Already reworked by GF before (21): Protection, Respiration, Depth Strider, Frost Walker, Soul Speed, Swift Sneak, Mending, Sharpness, Bane of Arthropods, Fire Aspect, Sweeping Edge, Looting, Power, Flame, Infinity, Piercing, Quick Charge, Riptide, Channeling, Wind Burst, Lure. Partly: Efficiency, Loyalty.
• **OPEN:** Breach and Thorns - Zeven says they were already reworked, but there is no record. Need the details.

7. Galaxium Fishing Rod (new item, decided)

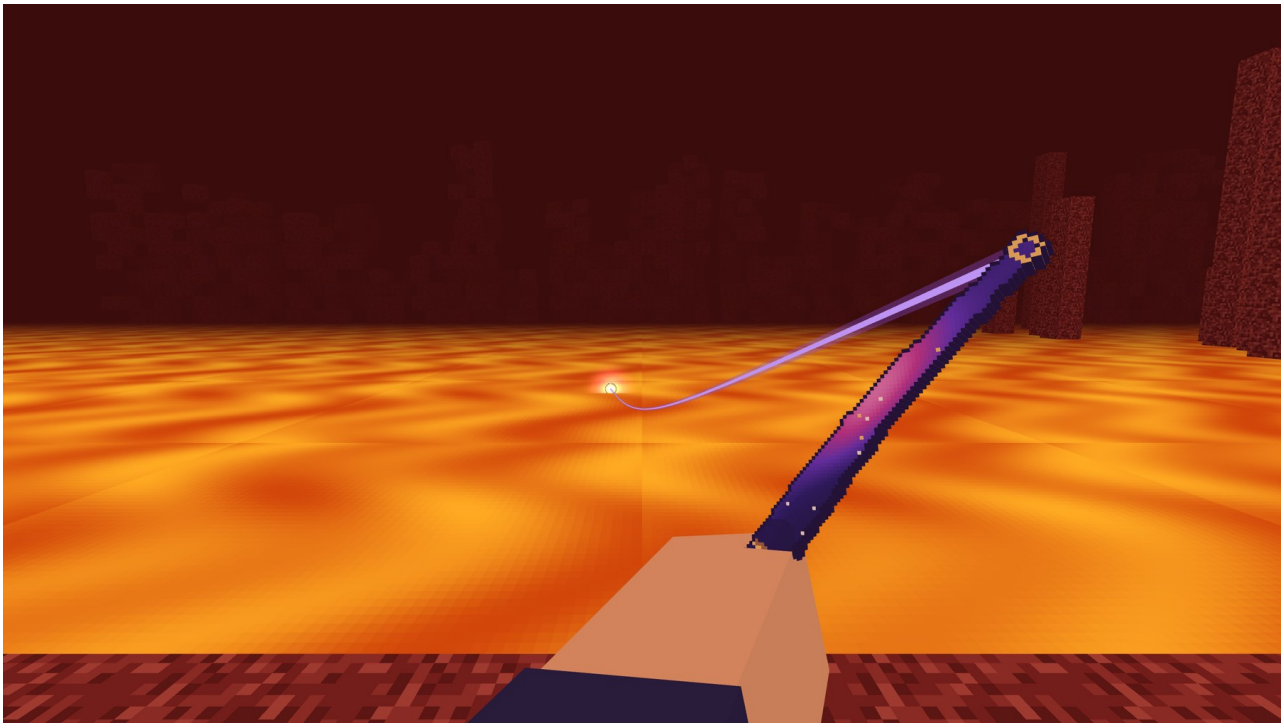
Part	Decision
Look	Concept B: glowing violet line + comet bobber, on the vanilla rod shape. 128px Galaxium texture, 60-frame seamless animation (item + cast).
Recipe	Smithing table: Fishing Rod + Galaxium Ingot + Galaxium Upgrade Template
Catching	2.5x faster; Lure stacks
Cast distance	5x
Durability	320 (5x a normal rod), can break
Protection	Fire/lava/explosion proof, never despawns; bobber floats on lava
Settings switch	Mob pull (strong: any pullable mob all the way to the player) OR grapple (bobber sticks to any block, pulls the player; no extra fall protection)
Lava fishing	With Luck of the Sea 50+, see section 6



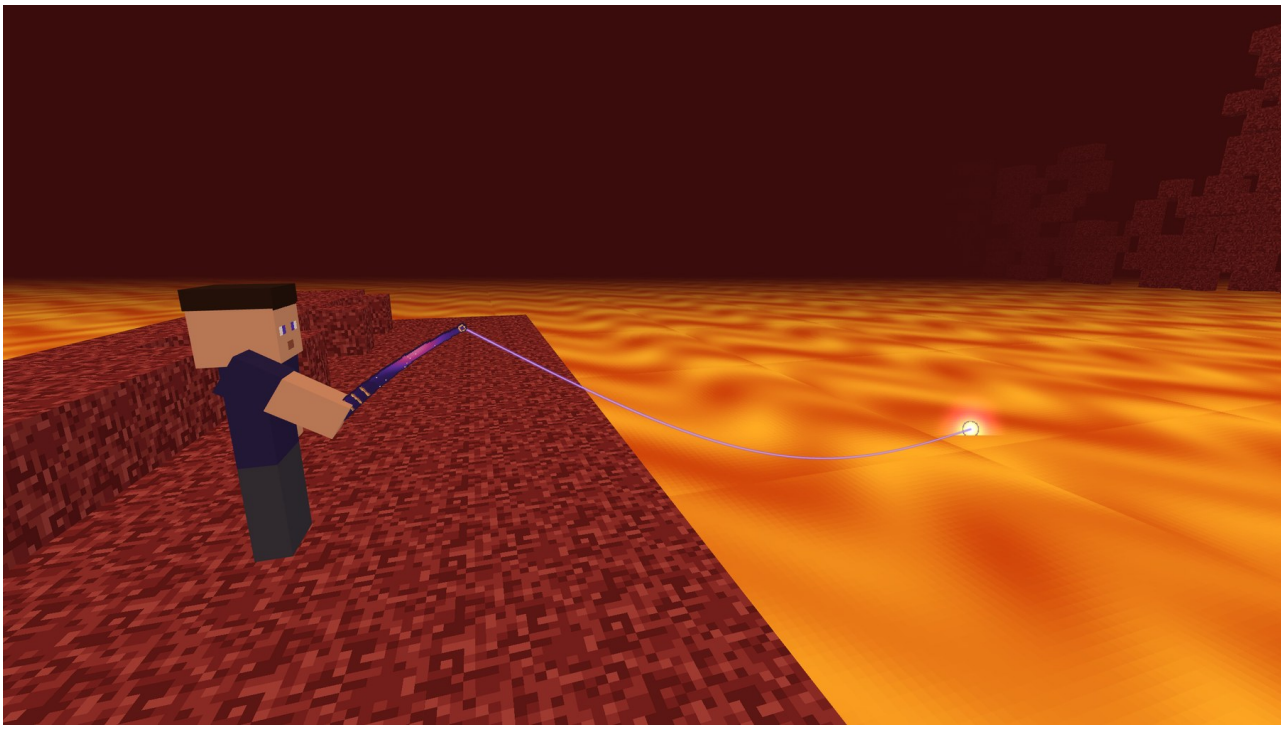
Chosen look: concept B on the vanilla rod shape (item, cast, bobber on lava, inventory size).



In hand, first person, holding it.



First person, fishing in Nether lava.



Third person, fishing in Nether lava. (3D previews, not game screenshots.)



The first 4 concepts (A-D). B was chosen.

8. Enchantment overview (before the reworks above)

• A separate PDF (GF-Enchantments.pdf) lists all 42 vanilla enchantments with vanilla max, 10x and 50x levels. Its max numbers come from the rule, not the code; it will be corrected after the jar audit.

9. Approved build plan

Rebuild the mod from the jar as real source (decompile with Vineflower into the NeoForge ModDevGradle project), prove the rebuilt jar equals the original, commit the source, then change it. Fallback: a small add-on mod that changes GF through mixins. Data (JSON) enchantment overrides where they match the decided behaviour exactly, code (events/mixins) elsewhere. Each phase = backup, change, verify, commit, one jar for Zeven to test.

Phase	Content
0 Setup + recovery	Jar upload, decompile, compile, prove equal to original, recreate resource/mixin/size checks.
1 Bug fix + audit	Fix Multishot cap; audit all 42 caps (show list before fixing); tree bone meal 8x/24x.
2 Level limit > 255	Mixins for saving/loading/sync/tooltips/'/enchant'; Nexus tier caps 10x/50x/75x/100x; test on a world copy.
3 Enchantment reworks	JSON: Knockback/Silk Touch reach, Feather Falling, Impaling, Punch on Galaxium Crossbow. Code: Smite, Aqua Affinity, Blast/Projectile/Fire Protection, Punch gravity, Multishot rain, Fortune smelt, Luck of the Sea lava loot. Every perk gets a toggle in the item settings menu.
4 Galaxium Fishing Rod	Item, lava-proof bobber, grapple/pull modes, 60-frame texture, models, recipe, lang.
5 Docs	Update PDF, text summary, Obsidian notes.

Verification: compile + checks every phase, jar < 30 MB, dedicated server starts, automated GameTests for the maths/loot/targeting, then Zeven's in-game checklist. Reports always say what is NOT yet tested in the game client.

10. Still open / to-do

• Upload genesisforge-1.0.0.jar (blocks everything).
• Breach and Thorns: what do they do now?
• Test T5: dispensers firing Galaxium mini-boss eggs.
• Luck of the Sea: better odds at higher levels - redesign once more loot items exist.
• Other GF ideas (not designed): Galaxium final boss (summoned after all 8 mini-bosses), 10th Omni Nexus tab, new Galaxium weapon or tool.